Errors reveal the depth of human reasoning

Measuring depth of deliberation from the structure of mistakes
in millions of real decisions

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Draft note. *Methods are final; Results are written as a pre-registered analysis plan. Quantities to fill after the runs are marked* [like this].

Abstract

How deeply a person deliberates before deciding is central to theories of bounded rationality, yet it is rarely measurable in natural behaviour. Here we show that the depth of human reasoning can be read directly from the structure of the mistakes people make. We use competitive chess as a natural laboratory in which millions of real, skill-graded decisions are scored against an authoritative standard at progressively deeper levels of analysis. Players of different skill differ little on the moves they get right but are sharply separated by the depth at which their errors are exposed: stronger players are misled only by deeper traps, a quantity we term the *depth of satisficing*. This depth rises smoothly with skill, collapses under time pressure, and differs across phases of play, and it can be estimated for an individual to yield an interpretable cognitive profile. A predictive model incorporating inferred depth substantially outperforms a state-of-the-art behavioural model on slow, deliberative play—where deep reasoning matters most—while adding little where decisions are fast. These results indicate that people commit to the best option found at the depth of search they reach, and that what someone gets wrong, more than what they get right, reveals how they think.

1 Introduction

A long tradition in the decision sciences holds that people do not optimise but *satisfice*: they search through options until one crosses an acceptability threshold, then stop and choose (Simon, 1956). Allied ideas—bounded rationality, level-k reasoning, and the contrast between fast intuitive and slow deliberative thought (Kahneman and Tversky, 1979; Stahl and Wilson, 1995)—all turn on a hidden quantity: how far a decision maker looks before committing. That quantity has been inferred almost exclusively from small staged experiments in artificial settings. Whether the regularities found there govern real, repeated, high-stakes choice is hard to test, because natural decisions rarely come with a ground-truth measure of how good each option was, let alone how good it *looked* at successive depths of consideration.

Chess supplies exactly this. Millions of decisions by players of precisely known skill are recorded in real competition, and each can be scored against an engine far stronger than any human—at every depth of an iterative-deepening search. The engine's evaluation of a move *changes with depth*: a move that looks strong shallow may be refuted only deeper, and a move that looks weak at first may reveal its worth later. This depth-resolved record lets us ask, of every human mistake, *at what depth the flaw becomes visible*, and therefore how deep the player must have looked.

Our central construct is the **depth of satisficing**: the depth at which the move a player chose stops looking better than the move they should have chosen. We show that this is the dimension along which skill is most legible. Players of widely different ratings are nearly indistinguishable on decisions they get right; the information separating them is concentrated in the small fraction of moves on which they err, and specifically in the depth at which those errors surface.

We make three advances over prior chess-error and engine-depth work (Regan and Haworth, 2011; Biswas and Regan, 2015; McIlroy-Young et al., 2020). First, we replace descriptive curve-fitting with a generative model in which the player is a latent distribution over search depths, so depth of satisficing is a posterior quantity with stated uncertainty. Second, we validate that engine depth is a meaningful proxy for human reasoning depth in three independent ways—predictive, convergent, and differential—rather than assuming it. Third, we use inferred depth to build interpretable individual profiles and to predict held-out human moves, outperforming a strong state-only behavioural model precisely where theory says deep reasoning should matter, but not elsewhere; this is notable given that strong searchless models exist (Ruoss et al., 2024).

2 Results

Skill lives in the errors, not the successes. Across [N] decisions spanning ratings 1200–2600+ under classical time controls, the average error of the played move and of the engine's preferred move are nearly indistinguishable at shallow depth and diverge only as depth increases (Fig. 1). On outright blunders skill is weakly informative; on "swing-down" decisions—played move good shallow, overtaken later—skill is strongly informative, carrying [X]× the rating information per decision of the remaining moves.

Depth of satisficing rises with skill and collapses under time pressure. The posterior depth of satisficing increases monotonically with rating ([$d \approx$] at the lowest band to [$d \approx$] at the highest), with non-overlapping 95% credible intervals (Fig. 2a). Within players, it falls by [$\Delta d \approx$] plies on moves played under time pressure (Fig. 2b), at every rating level—a within-subject effect the staged literature can only approximate.

Inferred depth tracks real thinking time. Inferred depth, fit without clock data, correlates with the actual time spent on the move (Spearman $\rho =$[..], held out; Fig. 3), strongest in the middlegame—convergent evidence that the construct reflects deliberation.

A depth-aware model predicts human moves better, only where depth should matter. A fusion of a state-only model (Maia-3) and a search-only mixture-over-depths model improves held-out cross-entropy by [Δ nats] and top-1 match by [Δ%]. The gain is differential and pre-registered: large on high-swing positions and classical play, near zero on low-swing positions and blitz (Fig. 4; interaction [β, $p =$]). A state-only model cannot produce this pattern.

Individual profiles separate expert styles. A four-component profile (depth of satisficing, trap-susceptibility, deep-discovery rate, time-elasticity) recovers rating by nested cross-validation (held-out $R^2 =$[..]) without per-phase rescaling, and resolves a known difficulty: a one-dimensional score misranks two world champions, whereas the profile attributes their strength to different axes and no longer inverts their ratings (Fig. 5).

The instrument recovers known depth. On agents that satisfice at fixed, known depths, the model recovers each planted depth (mean absolute error [..] plies; Fig. 6), confirming it measures depth rather than position difficulty.

3 Discussion

The errors people make are usually treated as noise. We find the opposite: where success is widely shared, error carries the informative variation, and its fine structure—the depth at which a flaw becomes visible—is a measurable signature of how a person reasons. Depth of satisficing scales with expertise, contracts under time pressure, and powers prediction exactly in the slow regime where depth can vary. This supports the view that people commit to the best option found at the plateau of search they reach rather than integrating across the whole trajectory, consistent with level-k choices being largely independent of higher-order belief tails (Strzalecki, 2014). The approach generalises to any domain with skill-graded decisions and a depth-resolved value standard; we demonstrate one transfer (Methods). A constructive corollary: an agent that estimates a person's depth of satisficing could tailor problems just beyond it. *Limitations:* engine depth is a proxy for human selective search, validated but not isomorphic; chess is adversarial and highly trained, so the generality claim rests on the transfer experiments.

4 Methods

Data. Current online games from the Lichess open database, sampled across rating bands (200-point bins, 1200–2600+) and time classes (bullet/blitz = fast, rapid/classical = slow), with per-move clock times retained. An elite subset (world champions) supports the profile case study. Synthetic agents (below) provide ground truth.

Engine analysis and value scaling. Positions are analysed with Stockfish 18 in Multi-PV mode, recording each candidate move m_i's evaluation $v_{i,d}$ at every depth $d = 1 \dots D$ (D=22, K=16). The candidate set is the union of top-K across all depths so trap moves keep a full trajectory; a played move outside top-K is re-queried alone. With `UCI_ShowWDL`, the engine reports win/draw/loss probabilities directly, so the win probability $P_{i,d} = (W+0.5D)/1000$ is read off per move per depth—no logistic fit. Depth-d regret is

$$\delta_{i,d} = \max_j P_{j,d} - P_{i,d} \ \geq \ 0. \tag{1}$$

State-tower features. The human-pattern prior $q_i(p, c)$ is the Maia-3 policy at the side-to-move's rating, read from the policy head as a full distribution over legal moves.

Latent-depth choice model. A player who has effectively searched to depth d chooses by a soft-max over depth-d regret fused with the pattern prior as a product of experts,

$$P(\text{move}{=}i \mid d, p, c) = \frac{\exp(-\beta\, \delta_{i,d} + \alpha\, \log q_i)}{\sum_j \exp(-\beta\, \delta_{j,d} + \alpha\, \log q_j)}. \tag{2}$$

Effective depth is latent with distribution $\pi_d(p, c) = P(\text{searched to } d \mid p, c)$, emitted by a network over a coarse depth grid conditioned on context c (rating, time class, clock, move number, total swing). The move likelihood marginalises depth,

$$P(\text{move}{=}i \mid p, c) = \sum_d \pi_d(p, c)\, P(\text{move}{=}i \mid d, p, c). \tag{3}$$

Parameters are fit by minimising the negative log-likelihood of the played move y, $\mathcal{L} = -\log \sum_d \pi_d P(y \mid d)$, with an entropy penalty on π for identifiability and β tied globally; data are split by player.

Depth of satisficing. The posterior over effective depth given the observed move,

$$r_d = P(d \mid y, p, c) = \frac{\pi_d \, P(y \mid d)}{\sum_{d'} \pi_{d'} \, P(y \mid d')}, \qquad \hat{d} = \sum_d d \, r_d, \tag{4}$$

gives a per-decision depth with uncertainty; player/condition estimates use a hierarchical model with players as random effects and bootstrap/posterior credible intervals.

Profiles and rating recovery. Per player: mean $\hat{d}$, trap-susceptibility (swing-down rate weighted by pre-refutation attractiveness), deep-discovery rate, and time-elasticity (slope of $\hat{d}$ vs. clock). Rating is predicted from the profile by ridge regression under nested cross-validation. The champion comparison reports components separately.

Model comparison and the differential test. State-only (Maia-3), search-only, and fusion predictors are compared on held-out cross-entropy and top-1/3 match. The pre-registered hypothesis is an interaction: the fusion's gain over state-only is larger for classical than blitz and for high-swing than low-swing strata, tested in a mixed-effects model with players and positions as random effects.

Convergent and recovery validation. Convergent: correlate per-decision $\hat{d}$ (fit without clocks) with observed time-on-move, held out, by phase. Recovery: agents that play a soft-max over fixed-depth-d_{plant} evaluations generate games; the pipeline's $\hat{d}$ is compared to d_{plant}.

Generality transfer. The estimation is repeated in Go using a depth-resolved value network in place of the chess engine, testing whether depth of satisficing again rises with rank.

Reproducibility. Stockfish 18 (net [name]), Threads [N], Hash [MB], D=22, K=16; Maia-3 `maia3-79m` (commit [hash]); Lichess month [YYYY-MM]; sampling seed fixed. Strata and the primary interaction are registered before held-out evaluation.

Figure 1: Average error of played vs. engine move across depth, by rating and swing class.

Figure 2: (a) Depth of satisficing vs. rating with CIs. (b) Within-player collapse under time pressure.

Figure 3: Inferred depth vs. observed time-on-move.

Figure 4: Held-out prediction gain of fusion over state-only, by time control × swing.

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Figure 5: Champion profiles: multi-axis decomposition and rating recovery without inversion.

Figure 6: Synthetic-agent recovery: estimated vs. planted depth.